

# Project C Thermal Test Analysis Report

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## 1 Introduction

The three main ways that heat can be transferred are conduction, convection and radiation. In space, where there is a near-vacuum environment, there are no means get rid of excess heat through convection to the surroundings, and spacecrafts tend to utilize thermal radiation as a passive thermal control system. Different materials absorb or emit heat differently depending on their absorptivity and emissivity. In this lab we analyze these characteristics for four separate materials using the “Space Sauna” facility of the KTH Space Technology Laboratory. The facility is a thermal vacuum chamber, where pressures below  $10^{-2}$  mbar [1] may be reached while also maintaining control of the temperature of the chambers internal walls, which are called *shroud*. Through the vacuum process the heat transfer through convection gets diminished and the only factors at work are radiation (and thermal conduction through the suspension wires and cables).

## 2 Methodology

Four test objects with different surface coatings/treatments were chosen, which are shown in fig. 1, with the known characteristics detailed in table 1.

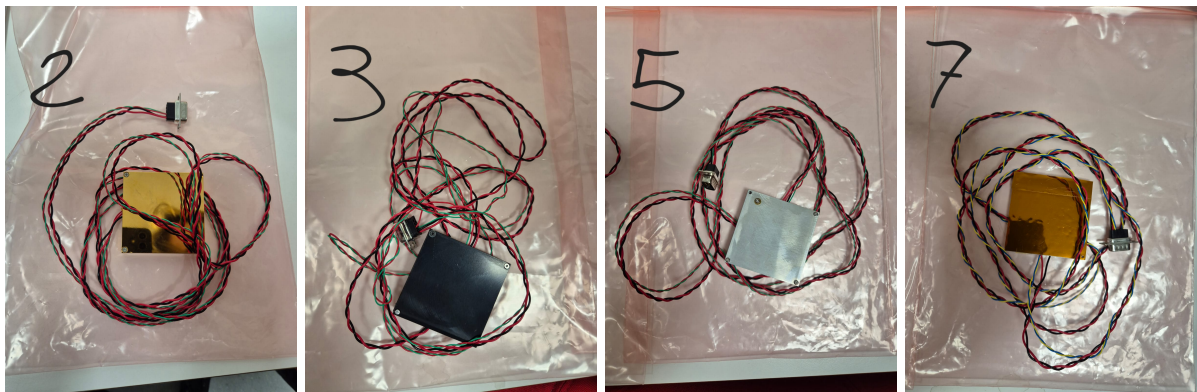


Figure 1: The test objects with surface coatings. Material of the coatings are listed in table 1.

| Surface coatings           | Mass (g) | Surface area (m <sup>2</sup> ) | Reference number |
|----------------------------|----------|--------------------------------|------------------|
| Gold plating (0.5 $\mu$ m) | 95       | 0.01105                        | 2                |
| Anodisation (black)        | 95       | 0.01105                        | 3                |
| SurTec 650 (3-valent Cr)   | 103      | 0.01105                        | 5                |
| Astral ATU2230             | 99       | 0.01105                        | 7                |

Table 1: Surface coating data

Each test object consisted of two aluminium plates, screwed together into a 65x65x10 mm<sup>3</sup> parallelepiped. A heater (Baker Calesco 50x50 mm<sup>2</sup> Polyimide heater) was fixed between the two plates in every sample. The test objects were cleaned using Isopropanol and thermal couplings were attached to each material. Subsequently, the samples were mounted on thin steel wires inside the vacuum chamber: a LUXEON MX high-power LED emitter was mounted furthest in, followed by the "Anodisation", "Gold plating", "Astral ATU2230" and "SurTec 650", in that order. The distance between the LED emitter and the first material was 12 cm, and the distance from the test objects to the walls was approximately 60 cm horizontally. A demonstration of the setup within the "Space Sauna" is shown in fig. 2, and the setup outside of the vacuum chamber is shown in fig. 3. Inside the cleanroom area of the lab facility, proper cleanroom clothing was worn at all times including disposable gloves, hair nets and lab coats.

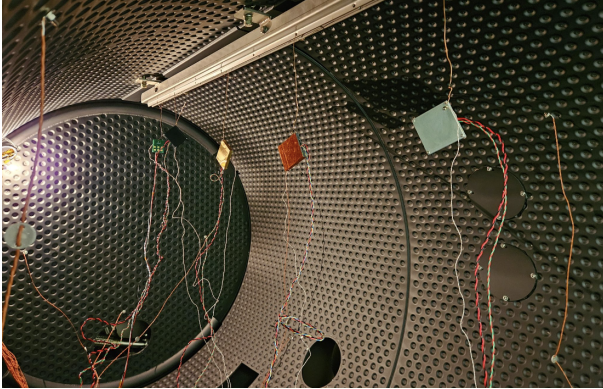


Figure 2: The setup inside the Space Sauna, demonstrating the suspension of the objects, and the LED light furthest in.

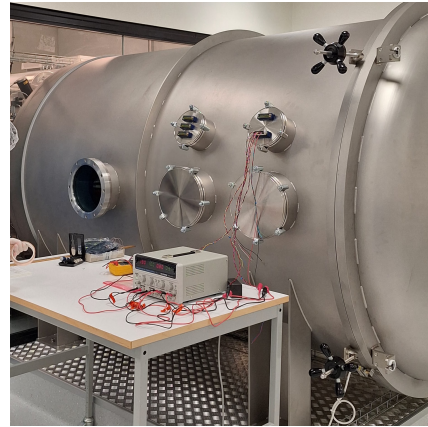


Figure 3: Setup of the power supply to the Space Sauna from which readings were taken.

The heaters were operated by the GPS-4303 power supply as pictured in fig. 3, which allowed control of the heating power through adjustable voltage and current for each channel. The voltages were tested to ensure that the thermal couplings were correctly attached, and the vacuum chamber was closed and began to pump out the pressure. Through trial and error, the voltages and currents were set for the separate channels to try and achieve a thermal steady state for the materials. During each trial, the voltage and current displayed on the power supply as well as the time of change was recorded. The air in the chamber was pumped out, and subsequently the shroud was cooled down to a temperature of  $-50^{\circ}$  C. The changing of voltage and current on the power supply continued after reaching  $-50^{\circ}$  C and until stable temperatures for each object were found. Due to the power supply utilizing 2 high voltage and 2 low voltage channels, this trial and error process involved changing channels between objects as well to satisfy the required power input for each object to reach thermal stability. The allowed span for heating the objects was up to approximately  $190^{\circ}$  C; above this, it was feared that the objects could encounter issues such as the surface coating flaking, which would not be optimal for the

chamber.

Later in the experiment, one of the channels, the one previously connected to the Astral object, was swapped with the LED light to test how much that heat contributed to the Anodized aluminum. Unfortunately, the LED light malfunctioned and its contribution could not be estimated (more about this in section 6). After the testing, the objects remained in the chamber overnight without any electrical power while the shroud slowly returned to room temperature, achieving smooth cooling curves for the objects.

To find the absorptivity,  $\alpha$ , and the emissivity,  $\varepsilon$  of the different test objects, the following equation was used

$$C \frac{dT}{dt} = P_{heat} + P_{abs} - P_e - P_{con} \quad (1)$$

where  $C$  is the heat capacity,  $P_{heat}$  is the electrical power supplied by the power supply,  $P_{abs}$  is the heat absorbed through radiation from the environment,  $P_e$  is the emitted heat from the object, and  $P_{con}$  is the conductive heat losses from suspension wires and cables.

In accordance with Kirchhoff's law, when the measurements are taken at the same wavelength,  $\alpha = \varepsilon$ , and since the system is in thermal equilibrium, the left hand side of Eq. eq. (1) is canceled, thus making it possible to estimate  $\varepsilon$  and  $\alpha$ . With those values at hand,  $C$  can be calculated by determining  $\frac{dT}{dt}$  from the recorded data and solving eq. (1) for  $C$ .

It is worth mentioning that  $\alpha = \varepsilon$  is strictly valid at a given wavelength and temperature. In the present experiment, the test objects and the chamber shroud emit thermal radiation at slightly different characteristic wavelengths due to their different temperatures. The use of this equation therefore represents a band-averaged approximation over the thermal infrared range. Given that both emissions lie in the mid-infrared and that surface properties vary only weakly over this range, this approximation is considered reasonable and its impact is small compared to other experimental uncertainties.

### 3 Preliminary calculations

The heat capacity of the chosen objects are given by

$$C(m) = mc_p \quad (2)$$

where  $c_p$  is the specific heat capacity of the test objects' material. As mentioned in section 2, the samples are primarily made of aluminium, which has a specific heat capacity of  $c_p = 897$  J/(kg K) [5]. Substituting into eq. (2) yields  $C(0.095) = 85$  J/K,  $C(0.099) = 89$  J/K and  $C(0.103) = 92$  J/K. Thus, the heat capacity of the objects are approximately  $C \approx 90$  J/K.

Assuming  $\varepsilon = 1$ , the maximum heat that can be emitted from the test objects at room temperature  $T = 20^\circ$  C and  $T = 190^\circ$  C is calculated as

$$P(T) = \sigma A_o T^4 = 5.67 \cdot 10^{-8} \cdot (2 \cdot 0.065^2 + 4 \cdot 0.065 \cdot 0.010) \cdot T^4, \quad (3)$$

where  $A_o$  is the object's emitting area, and  $\sigma$  is the Stefan-Boltzmann constant. This gives  $P(293) = 4.63$  W and  $P(463) = 28.84$  W.

The ambient radiation absorbed by the test objects, assuming  $\alpha = 1$  to evaluate the maximum absorbed heat, is calculated using the same formula as before, therefore at room temperature  $P(293) = 4.63$  and at  $-50^\circ$  C  $P(223) = 1.55$  W.

The fastest heating rate of the materials, if the heat absorption from the environment and the heat rejected through thermal radiation and conduction in cables and wires are disregarded, and

only the electrical power is considered, is computed as

$$\frac{dT}{dt} = \frac{P_{heat,max}}{C} \approx \frac{90}{90} = 1 \text{ K/s} \quad (4)$$

where  $P_{heat,max} = I_{max}V_{max} = 90 \text{ W}$ , which is the maximum power the power supply can produce [3].

To estimate the fastest possible cooling rate for objects at  $190^\circ \text{ C}$ , it is assumed that no external power is supplied and that  $\alpha = 0$  (disregarding radiative absorption from the walls), and  $\varepsilon = 1$ . The steel wire from which each object is suspended is taken to have a thermal conductivity of  $k = 20 \text{ W/mK}$ , a contact area of  $A_h = 2\pi rl = 2 \cdot \pi \cdot 0.3 \cdot 10 \cdot 10^{-6} \text{ m}^2$ , and a length of  $L = 0.3 \text{ m}$ . The contact area is assumed to be the surface area of the wire which can be assumed to have a cylindrical shape, since the suspension wire is thread in a small hole in the corner of the test objects. The temperature difference between the object ( $T_o = 190^\circ \text{ C}$ ) and the shroud ( $T_w = -50^\circ \text{ C}$ ) is therefore  $T_o - T_w = 240 \text{ K}$ . The conductive heat loss through the hangers is then

$$P_{hangers} = \frac{kA_h}{L}(T_o - T_w) \approx 0.31 \text{ W}. \quad (5)$$

Assume that the thermocouple and the four wires from the power supply are the same size and have the same contact area of  $A_p = 5 \cdot 2\pi rl = 5 \cdot 2\pi \cdot 0.2 \cdot 1 \cdot 10^{-6} \text{ m}^2$ , a length of  $L = 1.8 \text{ m}$ , and thermal conductivity as copper of  $400 \text{ W/mK}$ . The temperature is the same as the hangers, then

$$P_{cables} = \frac{kA_p}{L}(T_o - T_w) \approx 0.34 \text{ W} \quad (6)$$

The resulting fastest possible cooling rate is then estimated by also considering the radiated heat loss from the object, as

$$\frac{dT}{dt} = \frac{-\sigma A_o T_o^4 - P_{cables} - P_{hangers}}{C} = \frac{(-28.84 - 0.31 - 0.34) \text{ W}}{90 \text{ J/K}} = -0.33 \text{ K/s} \quad (7)$$

The purpose of these preliminary calculations are to state the assumptions that were made, as well as provide reference values for extreme cases.

## 4 Results

The evolution of temperature as a function of time for the different objects is shown in fig. 4. The TC37 curve corresponds to the LED light that was damaged due to an accidental power surge from the power supply at approximately 14:37:00 at which time TC40 was reconnected to the power supply.

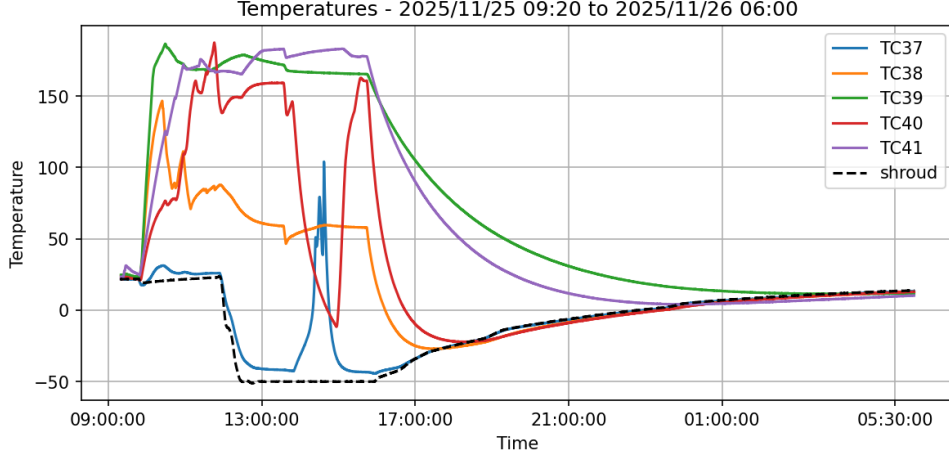


Figure 4: Full temperature evolution of the objects as a function of time.

The equilibrium points chosen for the calculations were at 13:33:47 for TC38 and TC40, and at 15:45:03 for TC39 and TC41. The corresponding objects to these channels along with their temperatures and power input are displayed in table 2. The equilibrium temperatures are not identical for all four objects due to limitations in the available heater power. As described in section 2, the power supply provides two high-power channels and two low-power channels, which restricted the maximum electrical power that could be delivered simultaneously. As a result, the objects connected to the low-power channels reached thermal equilibrium at lower temperatures than those connected to the high-power channels. The selected equilibrium points were nevertheless chosen such that the temperature slopes were close to zero, ensuring quasi-steady-state conditions suitable for emissivity and absorptivity calculations.

| Object                | Temperature (°C) | Power (W) |
|-----------------------|------------------|-----------|
| Anodized (TC38)       | 59.00            | 6.07      |
| Gold plating (TC39)   | 165.36           | 2.01      |
| Astral ATU2230 (TC40) | 159.28           | 17.95     |
| SurTec 650 (TC41)     | 177.82           | 2.96      |

Table 2: The temperatures of the different objects at their equilibrium points as well as the electrical power input.

Eliminating the left hand side in eq. (1) we can solve for  $\varepsilon = \alpha$  as

$$\varepsilon = \frac{P_{heat} - P_{con}}{\sigma A_o (T_o^4 - T_w^4)} . \quad (8)$$

Using the same values for the contact areas and lengths of the cables and suspension wires as in section 3, along with the temperatures and power input from table 2, the conductive heat loss for each object,  $P_{con} = P_{hangers} + P_{cables}$ , was calculated. The resulting values of  $\epsilon$  are presented in table 3.

| Object                | $\varepsilon = \alpha$ |
|-----------------------|------------------------|
| Anodized (TC38)       | 0.95                   |
| Gold plating (TC39)   | 0.07                   |
| Astral ATU2230 (TC40) | 0.86                   |
| SurTec 650 (TC41)     | 0.10                   |

Table 3: The absorptivity and emissivity of the different objects.

The heat capacity  $C$  can be calculated by finding a steep gradient in the curve and measuring the temperature drop over a known time step, as

$$C = \frac{P_{heat} + \sigma \epsilon A_o (T_w^4 - T_{o,avg}^4) - P_{con}}{\frac{dT}{dt}}, \quad (9)$$

where  $P_{heat} = 0$ , because we focus on times where the power was off, and  $T_{o,avg} = \frac{T_1 + T_2}{2}$  is the average temperature of the object during that time span. The time interval chosen for the temperature gradient was between 15:45:03 and 16:00:03, and the corresponding temperatures at those time steps are displayed in table 4, as well as the calculated  $C$  for each object.

| Object                | $T_1$ (°C) | $T_2$ (°C) | $C$ (J/K) |
|-----------------------|------------|------------|-----------|
| Anodized (TC38)       | 55.70      | 17.32      | 99.3      |
| Gold plating (TC39)   | 165.26     | 149.90     | 110.4     |
| Astral ATU2230 (TC40) | 159.07     | 78.86      | 132.2     |
| SurTec 650 (TC41)     | 177.21     | 152.26     | 95.5      |

Table 4: The temperatures at the begining and end of the time step used for  $\frac{dT}{dt}$ , and the heat capacity for each object.

The experimentally derived heat capacities show noticeable deviations from the theoretical value and also vary between the four test objects. Compared to the theoretical heat capacity estimated from material properties and geometry, the measured values differ by approximately 3.6–33% depending on the object. These discrepancies are discussed in more detail in section 6.

## 5 Sources of error and uncertainties

### 5.1 Conductive losses through hangers and cables

The derived emissivity and absorptivity values are affected by several experimental uncertainties arising from the heat balance shown in eq. (1), and from measurement errors in voltage, current, and temperature.

The suspension wire and the electrical/thermocouple wires provide conductive heat paths that contribute to  $P_{con}$ . Neglecting these conductive paths results in a systematic overestimate of the radiative power  $P_e$  and thus of the recovered emissivity  $\epsilon$ .

The losses for the hangers and the cables were estimated in eq. (5) and eq. (6), giving a total conductive loss

$$P_{con} \approx 0.65 \text{ W} \quad (10)$$

The net radiative heat loss appearing in the thermal balance is the net emission to the chamber shroud:

$$P_{net} = \sigma A_o (T_o^4 - T_w^4) \approx 10.4 \text{ W} \quad (11)$$

where  $A_o = 0.01105 \text{ m}^2$  is the total external area of the test object and a representative operating point ( $T_o = 190^\circ\text{C}$ ,  $T_w = -50^\circ\text{C}$ ) was chosen.

Thus conduction represents approximately

$$\frac{P_{\text{con}}}{P_{\text{net}}} \approx 6\% \implies \Delta\varepsilon_{\text{abs}} \approx 0.062 \quad (12)$$

of the apparent radiative heat loss. If conduction is neglected, the inferred emissivity is systematically overestimated by the additive amount  $\Delta\varepsilon_{\text{abs}}$  calculated in eq. (12). The relative bias depends on the true emissivity: for a low-emissivity surface ( $\varepsilon_{\text{true}} = 0.10$ ) this corresponds to a 62% relative error, whereas for  $\varepsilon_{\text{true}} = 0.50$  the relative error is 12%.

A conservative relative uncertainty of approximately 20% was assigned to  $P_{\text{con}}$ , reflecting uncertainties in wires diameter, thermal conductivity, contact conditions, and attachment temperature. This level is typical for unmeasured parasitic conduction paths in thermal-vacuum setups. This is equal to an uncertainty of approximately  $\pm 0.14 \text{ W}$ , equivalent to approximately  $\pm 1.4\%$  uncertainty in the recovered emissivity.

## 5.2 Electrical measurement uncertainties

The heater power  $P_{\text{heat}} = VI$  enters the heat balance explicitly, so errors in voltage or current measurement propagate directly into the recovered emissivity. Cable resistance causes the voltage displayed and outputted from the power supply to differ from the actual heater-pad voltage at the other end of the cable, leading to an overestimate of  $P_{\text{heat}}$  as the supply reading is used. Additional errors arise from: finite accuracy of the current readout, temperature-dependent heater resistance, asynchronous logging of  $V$ ,  $I$ , and  $T$ . These effects introduce both random and systematic errors in  $P_{\text{heat}}$ , and consequently in  $\varepsilon$ . Measuring the voltage directly across the heater terminals and synchronizing data acquisition substantially reduces this uncertainty.

## 5.3 Object geometry and thermal contact

Each test object consists of two bolted aluminium halves, introducing a thermal contact resistance at the interface. This can create temperature gradients between the thermocouple location and the radiating surfaces. If the object is assumed isothermal, such gradients lead to an incorrect estimate of  $P_e$  and therefore of  $\varepsilon$ . This effect is reduced near the steady state but may still shift the inferred emissivity and affects transient analyses used to extract heat capacity.

## 5.4 Radiative interaction between multiple objects

Multiple objects in the chamber exchange radiation with each other. An object can therefore absorb additional power from neighboring heaters, shifting  $P_{\text{abs}}$  and the apparent steady-state temperature. This adds uncertainty to both emissivity and absorptivity estimates. Experimentally, such coupling can be observed when modulating the heater power of one object produces measurable temperature changes in others. Minimising object-to-object view factors or powering objects individually helps reduce this effect.

This effect was tested after the LED malfunctioning, when the TC40 was reattached to the electric power so that the curve would be as steep as possible, in order to notice potential effects on other objects' stability. The conclusion from Fig. 5 was that the heating object had no effect on the thermal stability of the other objects, which all kept the same temperature throughout the re-heating of TC40.

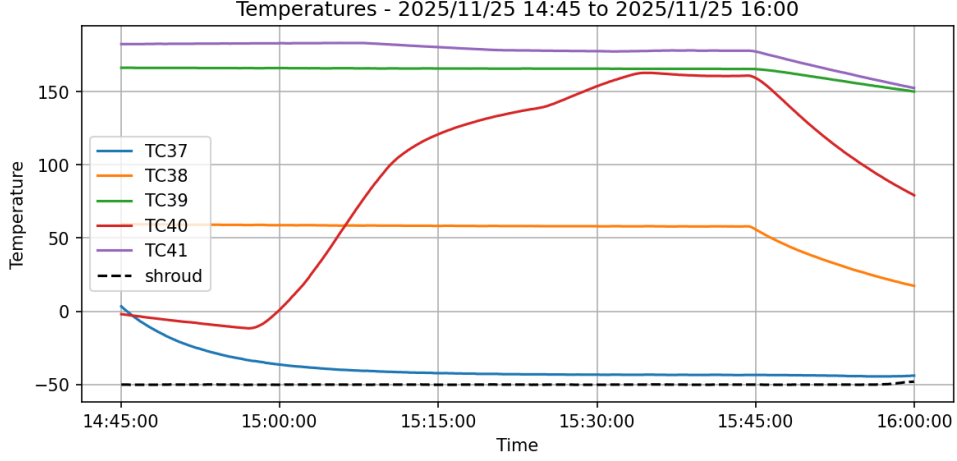


Figure 5: Fast heating of TC40 for radiative interactions test

The absence of a measurable thermal response in neighbouring objects can be explained by the radiative geometry of the setup. The mutual view factors between test objects are small, as most of each object’s radiative field of view is occupied by the chamber shroud rather than by other samples. Consequently, the dominant radiative exchange for each object occurs with the shroud, while object–object radiative coupling represents only a minor contribution to the overall heat balance. Any additional absorbed power from neighbouring objects is therefore expected to be small compared to the net radiative exchange with the shroud and below the sensitivity of the temperature measurements.

### 5.5 Heater resistance variations and interpolation errors

The heater resistance increases with temperature, changing the relationship between applied voltage, current, and true delivered power. If  $V(t)$  and  $I(t)$  are not recorded simultaneously, the reconstructed  $P_{\text{heat}}(t)$  contains interpolation errors. These errors influence both the steady-state determination of  $\epsilon$  and the transient fits used to estimate heat capacity. Using short logging intervals and measuring the voltage directly at the heater reduces these uncertainties.

## 6 Discussion

As shown in the previous sections, the heat capacities for all objects are not consistent with the expected values for the heat capacities, especially for the gold plating and the Astral ATU. This is most likely due to the fact that perfect equilibrium was never reached for the objects, so when the calculations assume equilibrium, the power is either too high or too low, resulting in a miscalculated value for the absorptivity and emissivity, which subsequently provides an error in the heat capacity. This is evident when comparing with the true values of emissivity. The true emissivity for gold plating is  $\epsilon = 0.05$ . Using this value while calculating  $C$  gives  $C = 90.8$ , which is very close to the expected value. The same applies for the rest of the coatings. The true values of  $\epsilon$  are represented in table 5.

| Object                   | True $\varepsilon$ |
|--------------------------|--------------------|
| Anodized (TC38)          | 0.88 [2]           |
| Gold plating (TC39)      | 0.05 [6]           |
| Astral ATU, Kapton 300HN | 0.57-0.78 [2]      |
| SurTec 650 (TC41)        | 0.06-0.13 [4]      |

Table 5: The documented emissivity of the different objects.

Given that the heat capacity is obtained from transient temperature slopes, the results are highly sensitive to measurement noise, time discretization, and uncertainty in the applied heater power. Small errors in temperature measurement or timing directly affect the estimated slope and therefore the extracted heat capacity. Another reason for these discrepancies is that the thermal model assumes the objects are perfectly isothermal, whereas in reality internal temperature gradients exist during transients, especially near heater locations and interfaces between the two aluminium halves. This leads to an effective heat capacity that differs from the lumped theoretical value.

As mentioned in section 5.1, the conductive heat loss is approximately 6% of the net radiative heat loss. This estimate is based on approximate values for the suspension wire dimensions, wire length, and thermal conductivity, and assumes an idealised cylindrical contact area between the wire and the object. In practice, the true contact geometry and attachment temperature are uncertain, which introduces an uncertainty in the estimated conductive heat loss. However, even when accounting for these uncertainties, conductive losses remain a secondary contribution compared to radiative exchange with the shroud. Reasonable variations in the assumed geometry or contact conditions change the conductive term by at most a few tenths of a watt, which corresponds to a small additive uncertainty in the recovered emissivity. The main conclusions regarding the dominant heat transfer mechanisms and the relative importance of conduction therefore remain unaffected.

The purpose of the LED experiment was to introduce an additional, controlled radiative heat input to the object, which would cause a measurable increase in equilibrium temperature or a change in the transient temperature slope. This change could then be used to estimate the surface absorptivity. In practice, the LED did not operate in a stable manner and required frequent adjustments of the power settings due to its steep current-voltage characteristics. As a result, the illumination periods were short and inconsistent. Because of the short time frames, the temperature response attributable to the LED could not be clearly distinguished from background fluctuations and other heat transfer mechanisms, such as radiation to the shroud and conductive losses through wires. Consequently, it was not possible to reliably isolate the effect of LED illumination on the thermal balance. This prevented a meaningful quantitative determination of absorptivity, even though the qualitative effect of additional radiative heating was understood.

## 7 Conclusion

In this experiment, the thermal properties of four distinct surface coatings were successfully characterized in a thermal vacuum environment. By analyzing the steady-state equilibrium temperatures and transient cool-down curves, we derived the emissivity ( $\varepsilon$ ) and heat capacity ( $C$ ) for Anodized Aluminum, Gold Plating, Astral ATU2230, and SurTec 650. The results yielded high emissivity values for the Anodized ( $\varepsilon \approx 0.95$ ) and Astral ( $\varepsilon \approx 0.86$ ) surfaces, consistent with their application as thermal radiators, while the Gold and SurTec surfaces exhibited low

emissivity ( $\varepsilon \approx 0.07 - 0.10$ ), confirming their effectiveness as thermal insulators.

Discrepancies were observed in the calculated heat capacities, particularly for the Gold plating, which were attributed to conductive heat losses through the suspension wires (estimated at  $\approx 6\%$  of the total heat budget) and deviations from perfect thermal equilibrium. Additionally, the malfunction of the LED emitter prevented the planned analysis of specific radiative interactions between active components.

Overall, the experiment demonstrates that steady-state thermal measurements can be used to estimate surface emissivity with reasonable accuracy, provided that conductive losses are properly accounted for. Although uncertainties in heat capacity estimation and the incomplete LED experiment limited the determination of absorptivity, the observed trends and order-of-magnitude results remain physically consistent. The dominant sources of uncertainty were identified and quantified where possible, showing that radiative exchange with the shroud governs the thermal behaviour of the objects, while conduction and object–object interaction play secondary roles.

Future iterations of this experiment should implement a 4-wire measurement setup to eliminate voltage drop errors in the heater cables and employ lower-conductivity suspension lines to minimize parasitic conduction. Furthermore, verifying the integrity of active components like the LED prior to vacuum pump-down would ensure a more complete dataset for radiative coupling analysis.

## 8 Group Contributions

The contributions of each group member to the project work, presentation, peer review, and final report are summarized in Table 6.

Table 6: Contribution of each group member.

| <b>Name</b>         | <b>Lab Work</b> | <b>Report</b> | <b>Presentation</b> | <b>Review</b> |
|---------------------|-----------------|---------------|---------------------|---------------|
| Brian Bertacco      | <i>Yes</i>      | <i>Yes</i>    | <i>Lab D</i>        | <i>Yes</i>    |
| Andreas Wangård     | <i>Yes</i>      | <i>Yes</i>    | <i>Lab C</i>        | <i>Yes</i>    |
| Jonas Planck        | <i>Yes</i>      | <i>Yes</i>    | <i>Lab C</i>        | <i>Yes</i>    |
| Samuele Cuzzi       | <i>Yes</i>      | <i>Yes</i>    | <i>Lab C</i>        | <i>Yes</i>    |
| Tobia De Benedictis | <i>Yes</i>      | <i>Yes</i>    | <i>Lab D</i>        | <i>Yes</i>    |
| Felice Venuti       | <i>Yes</i>      | <i>Yes</i>    | <i>Lab D</i>        | <i>Yes</i>    |

## References

- [1] Space sauna – termisk vakuumkammare. <https://www.kth.se/en/sci/centra/rymdcenter/rymdteknik/space-sauna-termisk-vakuumkammare-1.900572>, 2023. Accessed: 2025-12-05.
- [2] Emissivity values. [https://www.design1st.com/Design-Resource-Library/engineering\\_data/ThermalEmissivityValues.pdf](https://www.design1st.com/Design-Resource-Library/engineering_data/ThermalEmissivityValues.pdf), 2025. Accessed: 2025-12-05.
- [3] Gw instek gps-4303 power supply. <https://eleshop.eu/gw-instek-gps-4303.html>, 2025. Accessed: 2025-12-17.
- [4] M.Muccino. Next-generation lunar laser reflectors and applications for lunar gw detectors. <https://indico.cern.ch/event/1199289/contributions/5449031/attachments/2704224/4694181/Muccino%20-%20TAUP%202023.pdf>, 2023. Accessed: 2025-12-05.
- [5] Wikipedia contributors. Table of specific heat capacities. [https://en.wikipedia.org/wiki/Table\\_of\\_specific\\_heat\\_capacities](https://en.wikipedia.org/wiki/Table_of_specific_heat_capacities), 2025. Accessed: 2025-12-05.
- [6] G. Zhou and A. A. Mardon. Gold plating for lunar solar protection and temperature control. <https://www.lpi.usra.edu/meetings/leag2010/pdf/3019.pdf>, 2010. Accessed: 2025-12-05.